Seeing the Light at El Farol

A Look at the Most Important Problem in Complex Systems Theory

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THE WORLD OF EL FAROL

El Farol is a sawdust-on-the-floor, down-home-type of bar-restaurant on Canyon Road in Santa Fe, at which Irish music used to be played every Thursday evening. In an earlier era, Irish economist W. Brian Arthur of the Santa Fe Institute was fond of hoisting a pint or two on weekly outings to hear these memories of his youth. But he wasn't fond of doing so in the midst of a madhouse of pushing-and-shoving, shouting drinkers and lounge lizards. So Arthur's problem each Thursday was to decide whether the crowd at El Farol would be so large that the spiritual uplift he received from the music would be outweighed by the irritation of having to listen to the performance drowned in the shouts, laughter, and raucous conversation. Being a man of logical leanings, Arthur attacked the question of whether to attend or not in analytical terms. In the process he came to some striking conclusions about that bugaboo of the economic profession, rational expectations.

Assume there are 100 people in Santa Fe, each of whom, like Arthur, would like to listen to the music. But none of them wants to go if the bar is going to be too crowded. To be specific, suppose that all 100 people know the attendance at the bar over the past several weeks. For example, such a record might be... 44, 78, 56, 15, 23, 67, 84, 34, 45, 76, 40, 56, 23, and 35 attendees. Each individual then independently employs some prediction procedure to estimate how many people will appear at the bar on the coming Thursday evening. Typical predictors of this sort might be:

i. the same number as last week (35);
ii. a mirror image around 50 of last week's attendance (65);
iii. a rounded-up average of the past four weeks attendance (39);
iv. the same as two weeks ago (23).

Suppose each person decides independently to go to the bar if his or her prediction is that less than 60 people will go; otherwise, the person stays home. In order to make this prediction, every person has his or her own individual set of predictors and uses the currently most accurate one to forecast the coming week's attendance at El Farol. Once each person's forecast and decision to attend has been made, people converge on the bar, and the new attendance figure is published the next day in the newspaper. At this time, everyone updates the accuracies of all of the predictors in their particular set, and things continue for another round. This process creates what might be termed an "ecology" of predictors.

The problem faced by each person is then to forecast the attendance as accurately as possible, knowing that the actual attendance will be determined by the forecasts others make. This immediately leads to an "I-think-you-think-they-think..."-type of regress, a regress of a particularly nasty sort. For suppose that someone becomes con-

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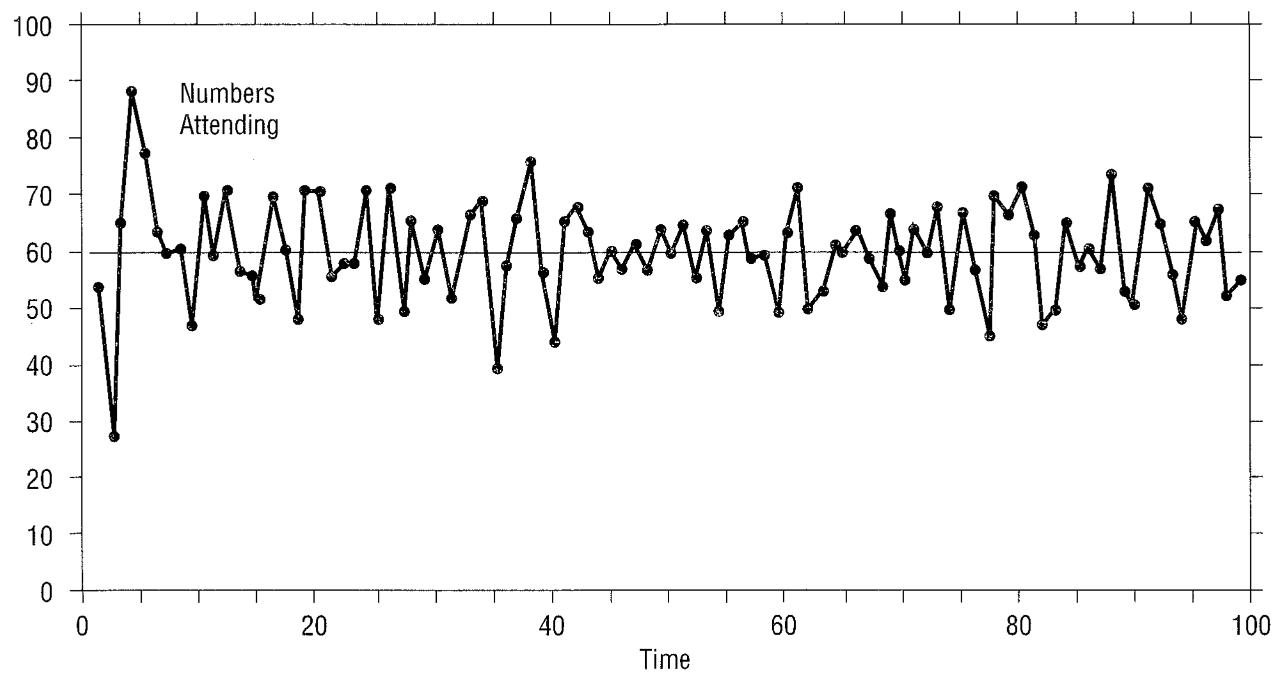
FIGURE 1



A simulated 100-week record of attendance at El Farol.

vinced that 87 people will attend. If this person assumes others are just as smart, then it's natural to assume that they will also see that 87 is a good forecast. But then they all stay home, negating the accuracy of that forecast! So no shared, or common, forecast can possibly be an accurate one; deductive logic fails. Hence, from a scientific point of view, the problem comes down to how to create a *theory* for how people decide whether or not to turn up at El Farol on Thursday evening and for the dynamics that these decisions induce.

A SILICON SURROGATE

It didn't take Arthur long to discover that it seems to be very difficult even to formulate a useful model of this decision process in conventional mathematical terms. So he decided to turn his computer loose on the problem and to create the "would-be" world of El Farol inside it, so he could study how electronic people would act in this situation [1]. What he wanted to do was look at how humans reason when the tools of deductive logic let them down. As an economist, Arthur's interest is in self-referential problems, situations where the forecasts made by economic agents act to *create* the world they are trying to forecast. Traditionally, economists look at such worlds using the idea of *rational expectations*. This view assumes homogeneous agents, who agree on the same forecasting model, and know that others know that others know that... they are using this forecast model. The classical view then asks which forecasting model would be consistent, on the average, with the outcome that it creates. But nobody asks how agents come up with this magical model. Moreover, if you let the agents differ in the models they use, you quickly run into a morass of technical—and conceptual—difficulties.

What Arthur's experiments showed is that if the predictors are not too simplistic, then the number of people who will attend fluctuates around an average level of 60. And, in fact, whatever threshold level Arthur chose, that level always seemed to turn out to be the average of the distribution describing the number of attendees. In addition, the computational experiments turned up an even more intriguing pattern—at least for mathematical systems theorists. The number of people going to the bar each week is a purely deterministic function of the individual predictions, which themselves are deterministic functions of the past number of attendees. This means that there is no inherently random factor dictating how many people actually turn up. Yet the outcome of the computational experiments suggests that the actual number going to hear the music in any week looks more like a random process than a deterministic function. The graph in Figure 1 shows a typical record of attendees for a 100-week period when the threshold level is 60.

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These experimental observations lead to a fairly definite and specific mathematical conjecture:

- Under suitable conditions (to be determined) on the set of predictors, the average number of people who actually go to the bar converges to the threshold value as the number of time periods becomes large.

System theorists might also like to ponder the related conjecture:

- Under the same set of suitable conditions on the sets of predictors, the time-series of attendance levels is a deterministic random process, i.e., it's "chaotic."

The major obstacle to a resolution of these conjectures is the simple fact that there currently exists no mathematical formalism within which to even meaningfully phrase these questions. I've just done it in a few words in everyday English, and Arthur created a world within which to experiment with these conjectures in a few lines of computer code. But to date, there is no mathematical formalism that we can use to actually *prove* (or disprove) these conjectures.

INDUCTIVE RATIONALITY

The upshot of the El Farol problem is that there exists no deductive chain of reasoning that will enable an individual to best decide whether to go to the bar or not. The implications of this fact for economies is clear. A forecast is only "fit," evolutionarily speaking, if it performs well in the world created by all the forecasts made by all the economic agents. Some predictors are mutually supporting while some others (like cycle predictors) are mutually negating. And, as the El Farol problem shows, common expectations are automatically negated. This diabolical twist leads to the view of an economy as a coevolutionary world in which new predictors are constantly being created so that agents can get a temporary advantage on their cohorts. But as others discover better ideas for prediction, these new rules for action are themselves soon undone—simply by their own success. And so it goes, new predictors constantly replacing the old, with the rule of survival in the marketplace leading to the Red Queen hypothesis of ecology: "You have to evolve as fast as you can just to stay in the game."

How do people really behave in situations like the El Farol problem? The answer is what Arthur terms "inductive rationality." Conventional economic wisdom dictates that agents can process the information available to them in a purely logical, deductive fashion to arrive at the best decision to take in any given situation. Here we see that the right thing to do—to go or not to go—depends on what everyone else is doing. But since no individual knows what everyone else is doing, all he or she can do is apply the predictor from the set of predictors that has worked best so far. The inductive part comes from recognizing that the right predictor to use this week may not be the one used the previous week; as new information comes in about the total attendance each week, people must reevaluate the effectiveness of their particular set of predictors, a process that may result in their changing what they regard as their "best" predictor at any given time. So agents continually shift from one predictor to another in a purely inductive fashion in an attempt to make the best possible decision about what to do. And this is exactly the way Arthur's surrogate music fans behave in his silicon world.

THE MOST IMPORTANT PROBLEM

Let me now advance the perhaps contentious claim that the key components constituting this El Farol problem show up in almost every complex, adaptive system—at least those involving intelligent, adaptive decision-makers, which certainly constitutes almost every problem arising in the social and behavioral areas—including those traditionally approached using the formalisms and tools of game theory. The natural corollary of this claim is that a decent mathematical formalism within which to describe and analyze the conjectures posed earlier about the El Farol problem would go a long way toward the creation of a workable scientific *theory* of these systems.

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To avoid confusion, let me give an example to make it as clear as possible what I mean by a "decent" mathematical formulation of the El Farol problem. Let S_n denote the set of prime numbers less than or equal to n. Now suppose we would like to count the number of elements in this set. To assure my place in history, here's a mathematical formulation that "solves" this currently unsolved problem. If $f(n)$ is this number the number of elements in the set S_n, then

$$f(n) = \sum_{S_n} 1.$$

That's it! And you can tell your friends you saw it here first. But you will obviously do nothing of the sort as this "formulation" is unacceptable in polite mathematical society for the simple reason that, despite its elegant appearance, it is nothing more than a restatement of the question. It does not provide a tool for calculating $f(n)$ that we didn't have before. So, for the sake of definiteness, what I mean by a "decent" formulation is one that offers a method for attacking the conjectures given above.

Here are the pieces of the El Farol puzzle that, in my view, essentially define what I take to mean by the term "complex, adaptive system":

- *Medium number of agents:* In the El Farol problem we have postulated 100 Irish music fans, each of whom acts independently in deciding to go or not go to the bar on Thursday evening. In contrast to simple systems, like superpower conflicts, which tend to involve a small number of interacting agents, or large systems like galaxies or containers of ideal gases, which have a large enough population of "agents" that we can use statistical means to study them, complex systems involve what one might call a "medium" numbers of agents. So just like Goldilocks's porridge, which was not too hot and not too cold, complex systems have a number of agents that is not too small and not too big, but just right to create interesting patterns of emergent behavior.

- *Intelligent and adaptive*: Not only are there a medium number of agents, these agents are intelligent and adaptive. This means that they make decisions on the basis of rules, like the predictors used by the patrons at El Farol, and that they are ready to modify their rules on the basis of new information that comes their way. Moreover, these agents can generate new rules that have never before been used, rather than being hemmed-in by having to choose from a set of pre-defined rules for action. This means that an "ecology" of rules emerges, one that continues to evolve as the system goes about its business.

- *Local information*: No single agent has access to what *all* the other agents are doing. At most, each agent gets information from a relatively small subset of the set of agents and then processes this "local" information to come to a decision as to how he or she will act. In the El Farol problem, the local information is as local as it gets, since each Irishman knows only what he or she is doing and has no information about the actions taken by any other agent in the system. This is an extreme case, however, and in most complex systems the agents are more like drivers in a road-traffic network or traders in a speculative market, each of whom has information about what at least a few of the other drivers or traders are up to.

"The powers of mind are everywhere ascendant over the brute force of things."

So these are the components of a complex, adaptive system: a medium-sized number of intelligent, adaptive agents interacting on the basis of local information. At present, there appear to be no known mathematical structures within which we can comfortably ask, let alone answer, natural mathematical questions about the El Farol situation like the conjectures posed above. This situation is completely analogous to that faced by gamblers in the 17th century, who sought a rational way to divide the stakes in a game of dice when the game had to be terminated prematurely by the appearance of the police (or, perhaps, the gamblers' wives). The description and analysis of that very definite real-world problem led Fermat and Pascal to create a mathematical formalism we now call probability theory. At present, complex systems theory still awaits its Pascal and Fermat.

Obviously, what was important to Fermat and Pascal was not the specific problem in dice-throwing, but rather the overall issue of how to characterize mathematically a general set of such problems. And the same is true for the El Farol situation. The problem itself is merely a cute little puzzle. But it is representative of a much broader class of problems that permeate almost every nook and cranny of the social and behavioral sciences. And it is this class of deep-reaching questions that a good formulation of the El Farol problem will help unlock. As philosopher George Gilder noted, "The central event of the twentieth century is the overthrow of matter. In technology, economics, and the politics of nations, wealth in the form of physical resources is steadily declining in value and significance. The powers of mind are everywhere ascendant over the brute force of things" [2]. And it is the development of a workable theory of complex systems that will be the capstone to this transition from the material to the informational. A good formulation of the El Farol problem would be a major step in this direction.

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